Student Name: Om Kasvala

Enrollment Number: 92410133005

Division/Semester: 3EK2 / Semester 6

Course Assignment: TASK 4 - Project Planning with Timeline (HCD)

Project Title: DeskBot: An AI-Powered Cloud Interactive Assistant

1. Prototyping Plan

To ensure a step-by-step engineering approach, the prototyping phase is divided into three distinct stages. It moves from basic hardware testing to AI software training, and finally to a fully assembled device.

Stage 1: Low-Fidelity Hardware & Breadboard Prototyping (Proto-1)

- **Objective:** Build the physical circuit on a breadboard to test the output screens (OLED expressions) and sounds (buzzer), as well as the touch sensor input, before connecting it to the internet.
- **Tools & Materials:** ESP8266 NodeMCU, two 0.96-inch OLED displays, TTP223 touch sensor, passive buzzer, breadboard, jumper wires, Arduino IDE, Adafruit_GFX libraries.
- **Due Date:** March 10, 2026

Stage 2: AI Model Training & Cloud Integration (Software Model)

- **Objective:** Collect sample images of good and bad posture to train/tune the AI classification logic. Once the AI can accurately detect a "slouch," integrate it with AWS IoT Core to securely send that alert to the breadboard.
- **Tools & Materials:** Python 3.10, OpenCV (for camera feed), MediaPipe (for extracting body points), Scikit-learn/Math logic (for training the posture classifier), AWS IoT Core (for secure MQTT message routing), VS Code.
- **Due Date:** March 20, 2026

Stage 3: High-Fidelity Physical Prototype (Final Assembly)

- **Objective:** Move the completed circuit off the breadboard into a user-friendly physical case for real-world testing on a desk.
- **Tools & Materials:** 3D Printer (PLA Filament) for the outer shell, soldering iron, perforated circuit board (perfboard), heat shrink tubing, standard 5V USB power supply.
- **Due Date:** April 5, 2026

2. Testing Plan

Following Human-Centered Design (HCD) principles, testing will focus on user interaction, emotional response, and behavioral changes, while simultaneously validating the full Edge-to-Cloud-to-Edge engineering pipeline.

Target Users: The testing cohort will consist of 5 individuals: University students and Work-From-Home (WFH) professionals who spend a minimum of 6 continuous hours per day at a desktop computer.

Phase 1: Usability & Environmental Testing (Software Model)

- **Testing Process:** Test the Python computer vision script's ability to accurately detect "slouching" across different users (varying heights and chair types) and under different environmental lighting conditions (daylight vs. low-light room).
- **Feedback Collection & Analysis Due Date:** March 25, 2026

Phase 2: System Integration & A/B Testing (AI, Cloud, and HCD Interface)

- **Testing Process:** Deploy the fully integrated system to test both user psychology and technical stability across the entire tech stack:

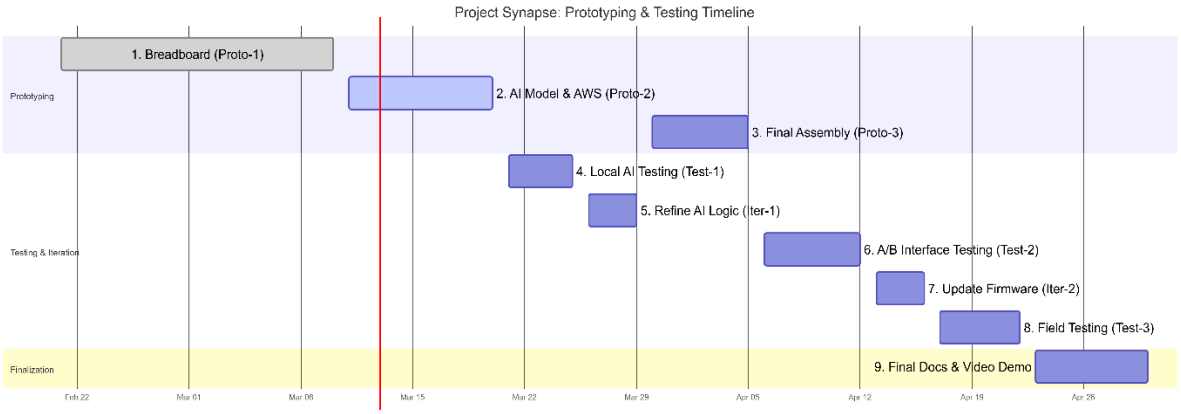
    - **HCD A/B Testing:** Evaluate user response to two different physical feedback profiles (Versions to measure effectiveness of face expressions of the bot).
    - **AI Vision Validation:** Monitor the Python/MediaPipe script's ability to accurately trigger these states without false positives while the user is actively typing or on calls.
    - **AWS IoT Cloud Performance:** Measure the MQTT message latency between the laptop AI and the physical robot to ensure sub-2-second delivery over standard Wi-Fi networks.
    - **IoT Hardware Stability:** Test the ESP8266 NodeMCU's ability to maintain a continuous cloud connection, process payloads, and render OLED animations without memory crashes.

- **Feedback Collection & Analysis Due Date:** April 12, 2026

Phase 3: Field Testing & Final Iteration (End-to-End)

- **Testing Process:** The fully assembled High-Fidelity DeskBot will be deployed on the target users' actual work desks for 3 full working days.
- **Feedback Collection & Analysis Due Date:** April 22, 2026
- *Note:* Data will be collected via exit interviews focusing on cloud reliability, device distraction levels, and overall impact on daily physical fatigue. The final firmware and AWS routing logic will be iterated based on these results.

Gantt chart:


Project Synapse: Prototyping & Testing Timeline
Prototyping
Testing & Iteration
Finalization
1. Breadboard (Proto-1)
2. AI Model & AWS (Proto-2)
3. Final Assembly (Proto-3)
4. Local AI Testing (Test-1)
5. Refine AI Logic (Iter-1)
6. A/B Interface Testing (Test-2)
7. Update Firmware (Iter-2)
8. Field Testing (Test-3)
9. Final Docs & Video Demo
Feb 22
Mar 01
Mar 08
Mar 15
Mar 22
Mar 29
Apr 05
Apr 12
Apr 19
Apr 26